GENERAL THETA FOUNDATIONS I: ACQUIRED GEOMETRY AND CAUSAL RESOURCE TRANSFER

QIAN QI

Abstract. For prepared causal experiments with a finite observable protocol projection and a history-uniform conditional calibration certificate, we prove a two-sided resource–risk transfer theorem. The full report may remain informative about an evolving hidden state; neither the current proxy nor the protocol mode is assumed to recover the posterior. Actual stopped occupation satisfies a killed-flow constraint. Conversely every finite feasible flow has a common causal realization with the same terminal measure and no larger cost, after unvisited circulations are removed. One live label per protocol mode and the stated output labels suffice. Mandatory incompatible actions give a matching lower count. For a metastable preparation–measurement protocol the resulting sharp law is the maximum of $(N/\kappa)^{-2/\alpha}$ and $(Nm/\kappa)^{-2/(d+\alpha)}$, with the physical waiting cost $\kappa = 1 + 1/p$ explicit, uniformly as the gate probability p tends to zero. Controlled hidden-state and noncommuting quantum instruments verify the same theorem, also on singular acquired supports. A killed-potential perturbation estimate controls numerical error and mean cost without a positive stop hazard at every call. Calibration, simulation deficiency, persistent labels, program description and temporary computation remain separately accounted for.

1. Introduction

A preparation, waiting period, measurement and terminal readout need not have interchangeable roles. A favorable observation may be reachable only after a long causal protocol. A law placed directly on favorable observations forgets the cost of reaching them; an observer that is handed the current protocol stage without retaining it forgets a different resource. The distinction persists when the protected physical state evolves and the full report remains informative.

We study the passage from positive prepared kernels and executable future tests to actually acquired task geometry, a finite-state causal implementation and a common decision risk. The central object is not a prescribed manifold or covariance. It is the terminal part of a finite-cost occupation flow, marked by an observable prediction proxy and its certified residual risk. Flow conservation records which reports can actually follow which protocol stages. A raw conditional calibration certificate compares this marked distortion with the protected future test after every legal history, including histories selected by earlier rejections.

Theorem 3.1 gives both an unrestricted-history lower bound and a common finite-state upper bound. The experiment has L observable protocol modes, whose marginal transition/report law is closed at the raw-kernel level. The remaining report can depend on the hidden state and on its posterior. The implementing program uses at most L live labels and m fixed terminal values. No approximate posterior, old report tape, external clock or persistent random seed is supplied. When every mode must occur and distinct modes require incompatible actions, L live labels are also necessary. In that case the same marked-flow profile bounds both architectures at exactly the same acquisition and terminal-label budgets.

The converse realization requires care. A finite solution of balance equations may include circulation in an unvisited recurrent class. Such circulation is not acquired information. We

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prove that its removal preserves *all terminal submeasures* and decreases every nonnegative additive resource. Thus the profile is not merely a formal optimization on reachable points, and no stationary-distribution or uniform-absorption assumption is inserted to identify it with an execution law.

The geometric consequence retains two independent obstructions. Low residual uncertainty of order s^2 may have mass of order s^α, while locating a representable prediction inside radius r has mass of order r^d. These actual masses give the two terms of the matching law. We then place this geometry behind a preparation gate of success probability p. Each measurement attempt costs exactly $1 + 1/p$ raw calls in mean, including rejected attempts. Theorem 5.2 preserves that factor rather than absorbing it into a constant. It therefore applies uniformly along metastable families with $p \downarrow 0$, provided the scaled budget remains feasible. Cantor product preparations give noninteger d in the same proof.

Two raw realizations retain information that the finite program does not store: a controlled hidden Markov sign and a coherently evolving qubit observed by noncommuting weak instruments. In both, the protocol projection is exact but the posterior is not recovered from that projection or the current proxy. The protected risk comparison, not posterior recovery, is the link used by the central theorem. A separate killed-potential estimate permits finite-precision implementation even in protocol modes where stopping is forbidden.

Relation to established methods. Occupation measures and stationary randomized realization are classical tools of constrained Markov decision theory [1, 2]. We do not claim those tools or conditional projection as new. Compactness of occupation measures is delicate outside the finite, bounded-cost setting [3]; the proof below uses bounded finite-mode counts and domination by fixed raw report measures, not a general compactness theorem for stopped strategic laws. Trimmed quantization and selective classification already optimize distortion or uncertainty under mass restrictions [6, 5]. The present theorem couples those restrictions to causal balance, unknown hidden dynamics, legal action signatures and a complete observer state count.

Value-directed belief compression [8] uses a task-sensitive representation; Bayes-risk quantization has a Bregman interpretation [9]. These facts do not alone provide the stopped all-history converse or pay for a protocol controller. Policy-conditional finite-memory approximation [10] compares values under a specified execution, whereas the lower bound here permits arbitrary legal history-dependent sensing and stopping. The full predictive quotient remains a separate, explicitly realized information object [7].

The one-protocol marked comparison, the reliability–resolution exponent, and the Gaussian all-history and two-state consequences are retained with complete proofs in the stopped-experiment companion [4]. The present contribution is their finite-protocol causal completion, including circulation removal, necessary live-state charges, a waiting-rate-uniform realization and drift-based resource perturbation. The raw projection and calibration hypotheses are substantive sufficient conditions. They are not a classification of arbitrary causal experiments or a claim to have learned an unknown kernel.

2. Prepared experiments and an observable protocol

All spaces are standard Borel. A preparation, a family of positive normalized causal kernels and a fixed bounded terminal audit are specified. A model parameter $\theta \in \Theta$ is fixed throughout a trajectory and is not available to the observer. The hidden physical state may include the entire microscopic past. The visible protocol mode belongs to $I = \{1, \dots, L\}$ and starts at the known mode i_0. Mode i admits a finite nonempty set of actions A_i. Costs $c_{ia} > 0$ are known and are paid for every call, including a rejected one. Write $c_* = \min_{i,a} c_{ia} > 0$.

The raw report is (J, Y, V): J is the next protocol mode, Y is a primary report and V is an additional informative report. Its joint law and the new hidden state are generated by the physical kernel, not by an observer. Let $\mathcal{H}_{t-1}$ include the complete visible past and all earlier private

randomizations. We require an *exact observable protocol projection*: for every model, every surviving legal history with mode i, and every selected action $a \in A_i$,

$$\mathrm{Law}_\theta(J_t, Y_t \mid \mathcal{H}_{t-1}, A_t = a) = P_{ia}(dj, dy). \tag{2.1}$$

The probability measures P_{ia} are known and common. No analogous independence is required for V_t or for the post-report hidden state. There need be no common report support across different modes or actions. Equation (2.1) is checked by marginalizing the original kernel on an invariant family of prepared physical laws; it is not inferred merely from the name "protocol mode." Every row used on a reachable mode is an actual conditional acquisition law of that physical experiment.

Stopping is allowed only on a specified Borel set $D_{ia} \subset I \times Y_{ia}$. It is decided after the whole current report; outside D_{ia} the observer must continue in the reported mode J_t. The audit $Z \in H$ is performed after commitment, with no intervening uncontrolled transition. Here $H \subset \mathbb{R}^k$ is compact and convex and contains all legal proxy and audit values. Projecting predictions onto H cannot increase squared loss. Denote the conditional audit mean and total variance, given the entire pre-audit history including (J, Y, V), by μ and v. These are regular conditional quantities, not probabilities assigned to continuous singletons.

Definition 2.1 (Conditional calibration)**.** Known Borel functions $f_{ia}(j, y) \in H$, $0 \le w_{ia}(j, y) \le W$, and constants $B_0 \ge 0$, $a_v > 0$, $A_v, a_b \ge 0$ satisfy, after every legal surviving history and every model, on the stop-allowed domain,

$$a_v w \le v - B_0 \le A_v w, \qquad \|\mu - f\|^2 \le a_b w. \tag{2.2}$$

The inequalities hold almost everywhere under the conditional full-report law. In particular they remain true after arbitrary selection using the additional report V. Fresh routing coins are conditionally independent of the audit. Values of f, w outside D_{ia} can be chosen arbitrarily within the displayed bounds.

For a raw report density λ, audit numerator U and audit squared-norm numerator T, relative to any local dominating full-report measure, the verification consists of

$$a_v w\lambda^2 \le T\lambda - \|U\|^2 - B_0\lambda^2 \le A_v w\lambda^2, \qquad \|U - \lambda f\|^2 \le a_b w\lambda^2.$$

These are imposed on a raw invariant family before any program or budget is chosen. At zero likelihood bounded tests force $U = T = 0$ and no posterior is assigned. Uniformity here concerns actually possible conditional histories, not a formal ambient state manifold. The proxy f and mode i are not asserted to be the complete predictive quotient.

A checkpoint uses arbitrary legal history-dependent randomized actions and stopping, with $1 \le \tau < \infty$ almost surely, and at commitment selects among at most m fixed values in H. An autonomous observer has disjoint live and terminal labels, starts at one fixed live label, chooses its action from that label alone, and routes using only its live label, current action and current report. Each terminal label has one fixed response. All rules are parameter independent; no retained seed, old report, real posterior or free external mode/clock register is allowed. A physical protocol mode is available for future action selection only insofar as its report has been encoded in the live label. Illegal actions and unauthorized stops are excluded, not offered as free probes. A visit to mode i means a live pre-call occurrence $I_{t-1} = i$ with $t \le \tau$; a next-mode mark on an accepted report is not by itself such a visit. Thus a positive-mass mode visit means a strictly positive expected call count there.

Let $R^{\Theta}_{\mathrm{cp}}(N, m)$ and $R^{\Theta}_{\mathrm{aut}}(N, M)$ be the infima of worst-model squared audit risk under

$$\sup_\theta \mathbb{E}_\theta \sum_{t=1}^{\tau} c_{I_{t-1}, A_t} \le N, \tag{2.3}$$

with M counting *all* live and terminal labels. Infeasible values are $+\infty$. The baseline B_0 is the same fixed task constant before optimization, not a difference of two separately optimized Bayes risks. Preparation and audit costs are added explicitly when a total physical budget is stated.

3. The causal acquired-flow theorem

For a dictionary $C = (r_1, \dots, r_m) \in H^m$ put

$$d_{ia,C}(j,y) = w_{ia}(j,y) + \min_{1\le b\le m} \|f_{ia}(j,y) - r_b\|^2.$$

The *acquired-flow profile* $\Psi_m(N)$ is the minimum of

$$\sum_{i,a} \int d_{ia,C}\, h_{ia}\, dP_{ia} \tag{3.1}$$

over $C \in H^m$, counts $x_{ia} \ge 0$ and Borel densities h_{ia} satisfying

$$0 \le h_{ia} \le x_{ia} \quad P_{ia}\text{-a.e.}, \qquad h_{ia} = 0 \quad \text{off } D_{ia}, \tag{3.2}$$

$$x_j := \sum_{a\in A_j} x_{ja} = \mathbf{1}_{\{j=i_0\}} + \sum_{i,a} \int_{\{j\}\times Y_{ia}} (x_{ia} - h_{ia})\, dP_{ia}, \tag{3.3}$$

$$\sum_{i,a} \int h_{ia}\, dP_{ia} = 1, \qquad \sum_{i,a} c_{ia} x_{ia} \le N. \tag{3.4}$$

The mass equation is redundant after summing the finite balance equations, but is displayed to prevent unnormalized terminal mass from being mistaken for a probability law. Empty feasible sets have value $+\infty$. The variables $h_{ia}P_{ia}$ are indexed by the *departing* mode and action and include the next reported mode.

Set

$$k_- := \left[\max\left\{2, \frac{1+2a_b}{a_v}\right\}\right]^{-1}, \qquad k_+ := \max\{2, A_v + 2a_b\}.$$

Theorem 3.1 (Acquired geometry with a paid causal protocol)**.** *Under* (2.1) *and Definition 2.1, for every finite N and $m \ge 1$,*

$$k_-\Psi_m(N) \le R^{\Theta}_{\mathrm{cp}}(N,m) - B_0 \le k_+\Psi_m(N). \tag{3.5}$$

For $M \ge L+1$,

$$k_-\Psi_{M-1}(N) \le R^{\Theta}_{\mathrm{aut}}(N,M) - B_0 \le k_+\Psi_{M-L}(N). \tag{3.6}$$

Every finite upper is implemented at the original budget by one common program with at most L live labels and the prescribed terminal labels. The profile minimum exists whenever its feasible set is nonempty. Every finite feasible flow admits a realization with exactly its terminal submeasures and no larger counts in any mode–action pair; unvisited circulation is not represented as acquired data.

If all L modes are visited before commitment with positive probability by every legal program under at least one common admitted model, and their action sets are pairwise disjoint, the first profile in (3.6) *can be replaced by $\Psi_{M-L}(N)$. Thus both architectures have matching profiles at the same paid terminal width. If additionally the exact conditional task identity*

$$\mathbb{E}[\|Z - r\|^2 \mid \textit{full pre-audit history}] = B_0 + w_{ia}(j,y) + \|f_{ia}(j,y) - r\|^2 \tag{3.7}$$

holds for every $r \in H$, then (3.5) *is an equality with coefficient one; under the mandatory-mode condition the autonomous equality with width $M - L$ also holds.*

The theorem does not optimize over a separate program for each hidden model. The lower bound is valid for every history-dependent procedure in each model; the upper is a single program that ignores V and uses only the common raw projection and certificate. It is the task comparison that permits this omission of information, not an assertion that V is uninformative.

Corollary 3.2 (Zero-risk boundary and several additive resources)**.** *For a feasible finite budget, baseline risk with m checkpoint values is attainable if and only if a feasible flow can be supported where $w = 0$ and f belongs to a dictionary of size m. The same equivalence holds for autonomous width $M - L$ under the mandatory-mode hypothesis. The theorem remains valid with finitely many nonnegative additive cost constraints, provided one constraint bounds the expected number of calls by a finite constant. All constraints retain their original units.*

Proof. Existence and positivity imply that $\Psi_m = 0$ exactly when a minimizing terminal measure has the asserted support. Apply the two positive constants in (3.5), or the sharpened version of (3.6). For several resources the compact count bound comes from the call-count constraint. The realization in the theorem decreases each count, so it satisfies every nonnegative cost constraint simultaneously. □

Local calibration without (3.3) is not a recursive-compressibility criterion. A low-risk report in a mode that cannot be entered from i_0 contributes no terminal mass in a feasible flow. Nor does the existence of the present finite observable projection follow from positivity of the original kernel; it is an explicit theorem-class assumption.

4. Proof of the central theorem

Lemma 4.1 (Conditional risk and actual occupation)**.** *At a permitted stop, for every legal committed prediction r,*

$$k_-\{w + \|f - r\|^2\} \le \mathbb{E}[\|Z - r\|^2 \mid \textit{pre-audit information}] - B_0 \le k_+\{w + \|f - r\|^2\}.$$

Every finite-cost history-dependent procedure induces a feasible flow in (3.2)–(3.4) *with its actual terminal submeasures.*

Proof. Conditional projection writes the middle expression as $v - B_0 + \|\mu - r\|^2$. The upper follows from $\|\mu - r\|^2 \le 2\|f - r\|^2 + 2a_b w$. For the lower,

$$w + \|f - r\|^2 \le (1 + 2a_b)w + 2\|\mu - r\|^2 \le k_-^{-1}\{v - B_0 + \|\mu - r\|^2\}.$$

Conditioning on a fresh response coin does not change the audit law.

For the second assertion fix a model and put

$$x_{ia} = \mathbb{E}\sum_{t\le\tau} \mathbf{1}_{\{I_{t-1}=i, A_t=a\}}, \quad \zeta_{ia}(B) = \mathbb{P}(I_{\tau-1} = i, A_\tau = a, (J_\tau, Y_\tau) \in B).$$

The count bound follows from $c_* > 0$. Predictability before the report and (2.1) give the full acquired report-count measure exactly as $x_{ia}P_{ia}$. The terminal count is dominated by that count, so $\zeta_{ia} = h_{ia}P_{ia}$ with $0 \le h_{ia} \le x_{ia}$ and the stop restriction gives its support. The number of visits to mode j is its initial visit plus the number of *continued* reports announcing j. Taking expectations proves (3.3). Almost-sure termination gives mass one. No conditional law after rejection has been substituted for a raw law. □

Lemma 4.2 (Finite-flow realization without phantom circulation)**.** *Every finite feasible (x, h) is realized in terminal law by a stationary randomized protocol program whose mode–action counts are at most x_{ia}.*

Proof. On a mode with $x_i > 0$ choose action a with probability $p_i(a) = x_{ia}/x_i$. If $x_{ia} > 0$, accept primary report (j, y) with probability $h_{ia}(j, y)/x_{ia}$; otherwise that action is never selected. Upon acceptance route to a nearest codepoint, with a fixed Borel tie rule. Upon rejection store mode j. Ignore the additional report V. On zero-count modes choose any legal row; they will not be reached from i_0.

The substochastic continuation matrix on positive-count modes is

$$Q_{ij} = \frac{1}{x_i} \sum_a \int_{\{j\}\times Y_{ia}} (x_{ia} - h_{ia})\, dP_{ia}. \tag{4.1}$$

The balance equation is $x = e_{i_0} + xQ$, with row-vector convention. Hence

$$\bar{x} := \sum_{n\geq 0} e_{i_0} Q^n \leq x.$$

Its finite total mass proves finite expected duration and almost-sure stopping for the constructed program. Equation (2.1) makes this continuation law the same for all hidden models.

The nonnegative residual $z = x - \bar{x}$ satisfies $z = zQ$. Summing its coordinates gives

$$\sum_i z_i(1 - Q_i\mathbf{1}) = 0.$$

Every summand is nonnegative. Thus $z_i = 0$ whenever mode i has positive acceptance probability; when its acceptance probability is zero, every h_{ia} is zero almost everywhere. Consequently the realized terminal density

$$\bar{x}_i p_i(a) \frac{h_{ia}}{x_{ia}} = \frac{\bar{x}_i}{x_i} h_{ia}$$

is exactly h_{ia} in all cases, with zero-count cases read as zero. Actual action counts are $\bar{x}_i p_i(a) \leq x_{ia}$. Unreachable closed classes may therefore have appeared in the finite balance witness, but they contribute neither terminal mass nor risk and are removed in its execution. □

Risk bounds and attainment. For a checkpoint dictionary C, Lemma 4.1 bounds the risk of its actual response below by the integral of $k_- d_{ia,C}$ against its actual terminal law. Taking the profile minimum, then the worst model and the infimum over common programs proves the first inequality of (3.5). Any autonomous program has at least one live label, so its terminal dictionary has at most $M - 1$ elements. This proves the first inequality of (3.6).

For existence, $0 \leq x_{ia} \leq N/c_*$ and $0 \leq h_{ia} \leq N/c_*$. Use the compact count box and the finite product of weak-star compact balls in $L^\infty(P_{ia})$. Pointwise inequalities, the permitted-stop restriction, balance and resource constraints are weak-star closed by testing against nonnegative L^1 functions and the finitely many mode indicators. The dictionary lies in compact H^m. Boundedness of f implies uniform convergence of the squared-distance functions when centers converge. The bounded objective is therefore jointly continuous in these topologies. A nonempty feasible set has a minimizer. This argument does not require compactness of a space of arbitrary infinite stopped trajectories.

Apply Lemma 4.2 to that minimizer. Its terminal submeasures are exactly the chosen $h_{ia}P_{ia}$, its costs are no greater, and conditional risk is at most $k_+ d_{ia,C}$. This proves the checkpoint upper and the autonomous upper with L live labels and m terminal labels. Under (3.7) the two conditional comparisons are identities, giving the exact checkpoint formula. □

Lemma 4.3 (Mandatory action signatures charge live states)**.** *Suppose b modes are visited with positive probability before commitment by every legal observer under one common model, and their permitted action sets are pairwise disjoint. Every such autonomous observer uses at least b live labels.*

Proof. At a live label s, the support of its fixed action row must lie in A_i whenever that label is actually used in mode i with positive probability. The same label cannot serve two of the stated modes: the support would lie in an empty intersection. Each mode has a positive-mass live label because the alphabet is finite. These labels are distinct. The argument does not use a free current-mode register, and an illegal action is not a permissible diagnostic experiment. □

Applying the lemma with $b = L$ gives the strengthened lower in Theorem 3.1; the exact autonomous conclusion follows from (3.7). More generally it gives $k_-\Psi_{M-b}$ whenever b mandatory incompatible modes are certified.

Example 4.4 (Two false relaxations)**.** First, take an initial mode that stops on its first report and a second, unreachable self-looping mode. The balance equations allow any finite positive occupation in the second mode, with zero terminal mass there. It is a circulation, not a second acquired population. Lemma 4.2 removes it exactly. Assigning terminal mass to it would violate balance.

Second, let a deterministic protocol require actions $a_1, \dots, a_L$ in that order, with disjoint singleton action sets and stopping allowed only after the last action. All reports can have constant task proxy and zero risk mark. A one-point geometric cover is then exact, but fewer than L live labels cannot execute the protocol. A cover or rowwise zero distortion alone is not a causal implementation.

5. Actual mass, global covering and metastable acquisition

We first record the geometric ingredient, with its proof, to distinguish it from the protocol completion. For a primary probability ν let $\sigma = (f, \sqrt{w})_{\#}\nu$ be its *actual* marked law. Suppose $f \in K \subset H$, $0 \le \sqrt{w} \le 1$, K compact, and

$$c_0 r^d s^\alpha \le \sigma(B(x,r) \times [0,s]) \le C_0 r^d s^\alpha \quad (x \in K,\ 0 < r \le r_0,\ 0 < s \le 1), \tag{5.1}$$

where d, α, c_0, C_0, r_0 are positive. This is not an assumption about the ambient hidden-state dimension or a postselected normalized stratum. Define

$$\mathcal{Q}_m(b) = \min_{C \in H^m,\ 0 \le h \le b,\ \int h\, d\nu = 1} \int (w + \min_j \|f - r_j\|^2) h\, d\nu, \qquad b \ge 1.$$

Proposition 5.1 (Reliability–resolution geometry)**.** *Under* (5.1)*, uniformly for $b \ge 1$ and $m \ge 1$,*

$$\mathcal{Q}_m(b) \asymp F_{d,\alpha}(b,m) := \max\{b^{-2/\alpha}, (bm)^{-2/(d+\alpha)}\}. \tag{5.2}$$

Both global covering and the requisite packing follow from the same acquired measure. This geometric proposition is the marked-mass result of the stopped-experiment companion [4]*, restated here to make the protocol theorem self-contained.*

Proof. Setting $s = 1$ gives d-regularity of the location marginal. Covering K by finitely many r_0-balls bounds the total low-mark mass by Cs^α. A selected density at most b leaves at least half its mass above $w = c^2 b^{-2/\alpha}$ for a fixed small c. This proves the first lower term.

The sublevel of distortion at most t^2 lies in m rectangles of radius t and height t. A center outside K with an intersecting ball can be replaced by an intersection point and the radius doubled. Its reference mass is at most $Cmt^{d+\alpha}$. Taking $t = c(bm)^{-1/(d+\alpha)}$ leaves at least half the selected mass outside, proving the other lower term.

A maximal r-separated subset is an r-cover and has at most Cr^{-d} points by the lower mass bound on disjoint $r/3$-balls. Conversely the upper mass bound guarantees k points separated by at least $ck^{-1/d}$. For large b select

$$k = \min\{m, \lfloor c_1 b^{d/\alpha} \rfloor\}, \qquad t = C_1 (bk)^{-1/(d+\alpha)},$$

with a zero floor replaced by one. Choose C_1 large enough for the k marked rectangles of radius and height t to have total mass at least $1/b$, and then c_1 sufficiently small to keep their location balls disjoint. Both choices can be made uniformly, since $k \le c_1 b^{d/\alpha}$. A constant density on their union gives terminal mass one with bound b and distortion at most $2t^2$. Since $k \asymp \min(m, b^{d/\alpha})$, this proves the upper. The compact bounded-budget interval is handled by the full law and one center; there F is bounded below by a positive constant. □

5.1. **A genuine preparation–measurement protocol.** There are two modes, G (gate) and A (acquisition), with disjoint action names. The preparation starts in G. A unit-cost gate call reports failure and returns to G with probability $1 - p$, or reports success and moves to A with probability p, where the protocol parameter $0 < p \le 1$ is known. Uniformity in p compares these known protocols; it does not presume that an unknown gate rate has been learned. Stopping at a

gate call is prohibited. At A, a unit-cost call produces a primary mark with law ν and an additional informative report. The observer either stops there or returns to G. Hidden physical dynamics are not reset by these protocol events. All actions within either phase, when present, have the same displayed primary projection and may control the hidden dynamics subject to the calibration certificate.

Theorem 5.2 (Waiting-rate-uniform causal resource law)**.** *For the preceding raw protocol let $\kappa(p) = 1 + 1/p$. The profile is feasible exactly for $N \geq \kappa(p)$, and*

$$\Psi_m(N) = \mathcal{Q}_m(N/\kappa(p)). \tag{5.3}$$

Under (5.1) *and calibration, for every $N \geq \kappa(p)$, $m \geq 1$ and $M \geq 3$,*

$$R^{\Theta}_{\mathrm{cp}}(N, m) - B_0 \asymp F_{d,\alpha}(N/\kappa(p), m), \tag{5.4}$$

$$R^{\Theta}_{\mathrm{aut}}(N, M) - B_0 \asymp F_{d,\alpha}(N/\kappa(p), M - 2). \tag{5.5}$$

The constants are independent of p, N, m, M and the hidden model, provided the calibration and geometry constants are fixed. For $M < 3$ no autonomous program is legal. Thus the result remains uniform along $p \downarrow 0$, including budgets N that vary with p; it does not assign a legal finite-cost procedure at $p = 0$.

Proof. Let x_G, x_A be the aggregate action counts. Gate termination is forbidden, so the acquired-flow equations give $x_A = px_G$. Total terminal mass one gives $x_A \geq 1$. Therefore

$$\mathbb{E}(\text{raw calls}) = x_G + x_A = \kappa(p)x_A.$$

This identity also follows directly for every history-dependent rule by predictable gate-success counting; no deterministic number of cycles or independent stopping time is presumed.

The terminal primary density is bounded by x_A and integrates to one. This gives the lower profile in (5.3). Conversely any density $0 \leq h \leq b = N/\kappa(p)$ of mass one is feasible with $x_A = b$ and $x_G = b/p$. At acquisition accept with probability h/b, otherwise return to the gate. The actual acceptance probability per acquisition is $1/b$, so its expected active-call count is b and its expected raw cost is exactly N. Both gate and acquisition modes occur before every commitment, and their action sets are disjoint. Lemma 4.3 therefore charges two live labels even when all task values coincide. Theorem 3.1 and Proposition 5.1 prove the claims. □

5.2. **Singular preparation and nonuniform waiting.** Let K be a finite product of intervals and middle-third Cantor sets with their uniform and fair Cantor probability measures. The location exponent is the sum of one for each interval and $\log 2/\log 3$ for each Cantor factor. Indeed at $3^{-(j+1)} < r \leq 3^{-j}$ a ball meets a bounded number of level-j cylinders, each of mass 2^{-j}, while a cylinder a fixed number of levels deeper through its center lies inside the ball. This proves both mass bounds for one Cantor coordinate. Product Euclidean balls are sandwiched between fixed rescalings of product rectangles, proving regularity on K. Independently draw T with $\mathbb{P}(T \leq s) = s^{\alpha}$ and use $w = T^2$. Then (5.1) holds for the *actual* prepared law.

The singular support and the metastable parameter p enter different places: d, α determine the geometric loss, while $\kappa(p)$ counts the physical calls needed to acquire that law. Neither is discarded by conditioning on gate success. At fixed feasible $N/\kappa(p)$, increasing the output dictionary cannot remove the reliability floor. The proof thus treats a nonuniformly waiting singular class, rather than appending a singular interface to a smooth theorem.

6. Two informative raw-kernel realizations

The protocol is the gate–acquisition instrument of Theorem 5.2. Let $Y \sim \nu_K$ and T have distribution $\mathbb{P}(T \leq s) = s^{\alpha}$, independently of each other and of the gate coins. The primary active report is (Y, T), with $f(Y, T) = Y$ and $w(Y, T) = T^2$. Choose one coordinate vector e_1 and a compact convex H containing $K + [-1, 1]e_1$. The complete report includes an informative sign

V. The raw driver projection is common; the full V likelihood is not required to be common or independent of the physical state.

Theorem 6.1 (Controlled hidden-state realization)**.** *At every raw call a hidden sign E first evolves by an action-selected transition matrix whose entries belong to $[q, 1-q]$, with fixed $0 < q \le 1/2$. The matrices may depend on an unknown fixed model. A gate call applies its independent Bernoulli protocol coin and permits no stop. At an active call, draw (Y, T) as specified, then report*

$$\mathbb{P}(V = v \mid E = e) = \frac{1 + \eta v e}{2}, \qquad 0 < \eta < 1.$$

After commitment audit $Z = Y + TEe_1$ without a new transition. Rejection keeps the hidden sign and returns the visible protocol to the gate. Then Theorems 3.1 and 5.2 hold uniformly over every admissible transition family and every $0 < p \le 1$. The law remains nonregenerative when its transitions have distinct rows.

Proof. All operations are normalized positive kernels. Before an active report, either sign has conditional probability at least q, for every surviving history and every chosen action. Let $\gamma = (1-\eta)/2$ and $a = q\gamma/(1-\gamma) > 0$. After observing V, both posterior sign probabilities are at least a. If b is the posterior sign mean, then

$$\mu = Y + Tbe_1, \qquad v = T^2(1-b^2), \qquad 4a(1-a)T^2 \le v \le T^2, \quad \|\mu - Y\|^2 \le T^2.$$

Thus Definition 2.1 holds with $B_0 = 0$, $a_v = 4a(1-a)$ and $A_v = a_b = 1$. These inequalities hold conditional on V, so the all-history lower allows stopping and action selection informed by it. The exact primary projection follows independently by marginalizing the raw kernel over the sign and V; gate and active laws are the displayed probabilities, at every surviving history. The true prior generally changes through rejections, and its effect on the posterior b does not disappear from the physical experiment.

For each fixed model, the finite sign instrument acts linearly on the two unnormalized sign masses. In the unknown-model interpretation these carriers are kept as a model-indexed family, or as a joint posterior only after a prior is specified; two masses alone do not encode uncertainty about a transition matrix. Include the reported Y, T, the protocol mode and legal action signature for a possible immediate audit. Report probabilities and the audit moments are executable functions of this carrier. Distinct beliefs are separated by future informative observations whenever a separating transition is admitted; degenerate directions are quotiented out, not declared identifiable. Report total variation is Lipschitz in the input sign law. Normalized active updates have denominators at least γ and are uniformly continuous on the compact belief interval. The observer does not store or numerically iterate this belief. Its exact stored protocol mode and the killed-flow realization supply the causal implementation. Proposition 5.1 verifies the global cover and mass bounds from the actual (Y, T) law. Constants depend on q, η but not on the gate rate p. □

Theorem 6.2 (Noncommuting quantum realization)**.** *Prepare a qubit in $I/2$. At each raw call apply the positivity-improving channel*

$$D_q(\rho) = (1-2q)\rho + q\ \mathrm{tr}(\rho) I, \qquad 0 < q \le 1/2,$$

followed by an action-selected unitary U_a. A gate call applies only the independent Bernoulli protocol coin and has no terminal readout. At an active call independently report (Y, T), use effects

$$K_{a,v} = \left(\frac{I + \eta v\, n_a \cdot \sigma}{2}\right)^{1/2}, \qquad \mathcal{I}_{a,v}(\rho) = K_{a,v} U_a D_q(\rho) U_a^* K_{a,v},$$

and report v. At commitment measure σ_z on the normalized current state, obtaining $B \in \{-1, 1\}$, and audit $Z = Y + TBe_1$. Rejection retains the quantum state and returns to the gate. Nonparallel directions n_a are allowed. The same two general theorems hold, with the same dependence on p, N, M, d, α as in the classical realization.

Proof. The channel and each instrument operation are completely positive, and $\sum_v K_{a,v}^2 = I$. Gate and active primary variables are physically sampled independently, giving (2.1). Put $\gamma = (1-\eta)/2$. Before a weak measurement, $U_a D_q(\rho) U_a^* \succeq qI$. Every effect has eigenvalues in $[\gamma, 1-\gamma]$, so its probability is in that interval and its normalized output obeys

$$\rho' \succeq \frac{q\gamma}{1-\gamma} I = aI.$$

The subsequent Pauli mean has absolute value at most $1-2a$. Its variance is therefore at least $4a(1-a)$ and at most one. Scaling by T and shifting by Y gives exactly the calibration inequalities proved above, now derived from the quantum raw kernel and Born audit, not imported from the classical theorem.

For each fixed model, positive operations on finite Hermitian matrices and executable Born effects give an unnormalized predictive representation. A uniform unknown-model quotient retains the model-indexed family, not a density matrix that silently encodes the true model. A tomographically separating probe family, when admitted, identifies density operators on the attained image; otherwise the corresponding redundant directions are factored out. The observable protocol signature is included in either case. Report probabilities are Lipschitz in trace norm, and normalization has denominator at least γ, proving report and predictive-update continuity. No infinite-dimensional operator closure is inferred. For $q < 1/2$, invertible weak effects can retain coherence after a rejection. Choosing axes x and z gives noncommuting effects. The physical qubit is not an available observer memory register: the implementation has precisely the charged classical live and terminal labels. Finally the actual primary location/reliability distribution verifies (5.1) independently of the quantum posterior. □

Weak measurement and its conditioning interpretation are standard [11]. The added protocol describes heralded preparation attempts followed by a measurement; every failure and subsequent attempt is paid. The examples deliberately verify a sufficient calibration class. The estimates do not become uniform as the hidden positivity floor q vanishes; their new uniformity is in the independently varying physical waiting rate p.

7. Finite precision without a one-call stopping floor

A protocol mode may prohibit stopping altogether. A lower bound on stop probability at every raw call is therefore the wrong implementation certificate. The appropriate object is a finite killed potential.

Theorem 7.1 (Killed-potential resource perturbation)**.** *For a finite live-label program, first discard live labels unreachable from the initial label under the reference execution. On the entire remaining live alphabet let Q be the substochastic continuation matrix, $c_i \geq c_* > 0$ a reference mean call cost and $b_i \geq 0$ a reference mean one-call terminal-loss contribution. Assume finite nonnegative vectors W, U on this alphabet satisfy*

$$W = c + QW, \qquad U = b + QU. \tag{7.1}$$

A possibly history-dependent numerical implementation remains action compatible, never enters a live label outside this certified alphabet, and for every surviving history with live label i satisfies

$$\mathbb{E}[W(S')\mathbf{1}_{\text{continue}} \mid \mathcal{H}] \leq (QW)_i + \rho c_i, \tag{7.2}$$

$$\mathbb{E}[\ell_{\text{term}}\mathbf{1}_{\text{stop}} + U(S')\mathbf{1}_{\text{continue}} \mid \mathcal{H}] \leq U_i + \eta c_i, \tag{7.3}$$

$$\mathbb{E}[c_{\text{actual}} \mid \mathcal{H}] \leq (1+\zeta) c_i, \tag{7.4}$$

where $0 \le \rho < 1$, $\eta, \zeta \ge 0$, and $\ell_{\mathrm{term}} \ge 0$ is the common bounded terminal loss. The inequalities are uniform in the hidden model. Then the implementation stops almost surely and

$$\mathbb{E}\sum_{t\le\tau} c_{S_{t-1}} \le \frac{W(i_0)}{1-\rho}, \qquad \mathbb{E}C_{\mathrm{actual}} \le \frac{(1+\zeta)W(i_0)}{1-\rho}, \tag{7.5}$$

$$R_{\mathrm{num}} \le U(i_0) + \frac{\eta W(i_0)}{1-\rho}. \tag{7.6}$$

No positive one-call stop probability, uniform mixing or comparison of discrete and continuous report laws in total variation is required.

Proof. Extend W, U by zero after commitment. By (7.1) and (7.2), the expected drop of W at a live call is at least $(1-\rho)c_i$. Sum up to $\tau \wedge n$, retain the nonnegative remaining potential, and let $n \to \infty$. Monotone convergence proves the first bound in (7.5). Since $c_i \ge c_*$, the expected number of calls is finite, so commitment occurs almost surely. Summing (7.4) proves the actual-cost bound. Summing (7.3) in the same way bounds the accumulated terminal-loss contribution by $U(i_0) + \eta\mathbb{E}\sum c_i$. There is exactly one terminal contribution almost surely, proving (7.6). Neither argument infers expected duration from a terminal-law distance. □

For a stationary approximation $(\widetilde{Q}, \widetilde{b}, \widetilde{c})$ it suffices to verify

$$(\widetilde{Q} - Q)W \le \rho c, \quad (\widetilde{Q} - Q)U + \widetilde{b} - b \le \eta c, \quad \widetilde{c} \le (1+\zeta)c$$

coordinatewise. These are finite certificate inequalities. Potentials are computed on the reachable transient part produced by Lemma 4.2; a phantom recurrent class is not inverted. When b majorizes the conditional terminal loss in every hidden model, the same construction gives valid U for that entire family.

Example 7.2 (A zero-hazard mode)**.** For the gate protocol with acceptance probability s per active call,

$$Q = \begin{pmatrix} 1-p & p \\ 1-s & 0 \end{pmatrix}, \qquad c = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad W_G = \frac{1+1/p}{s}, \quad W_A = W_G - \frac{1}{p}.$$

The gate stop hazard is zero, but the potentials are finite for every $p, s > 0$. At $p = 0$ no such finite potential exists. If gate coins and mode parsing are exact and each active decision changes by probability at most $e < s$ under the same raw primary report, direct cycle counting gives

$$\mathbb{E}C_{\mathrm{num}} \le \frac{\kappa(p)}{s-e}, \qquad R_{\mathrm{num}} \le \frac{b_A + L_\ell e}{s-e},$$

where b_A bounds the reference active-call terminal-loss contribution and $0 \le \ell \le L_\ell$. This follows by counting active attempts and the exactly $1/p$ mean gate calls preceding each attempt. The bound is not obtained by conditioning on successful gates for free.

Executable numerical certificates and accounting. The ideal theorems use Borel routing and real constants in the usual label model. A finite-description realization additionally requires computable centers, actions and acceptance cells, certified current-report evaluation, and a uniform bound on ambiguous boundary mass. Approximation couples the routing decisions on the *same physical report*; a discrete approximation to a continuous report is not close in total variation merely because its coordinates are close. Mode parsing and action legality must be preserved exactly. Certified arithmetic and boundary bounds then imply the finite inequalities above. One chooses a reference cost with enough slack to satisfy $(1+\zeta)W(i_0)/(1-\rho) \le N$; no budget violation is renamed approximation error.

For k-dimensional centers of b_r bits per coordinate, a mode/action table of description length L_A, and routing/coin descriptions of length L_γ, one explicit program bound is $O(kmb_r + L_A + L_\gamma + L\log(L+m))$. Persistent labels cost $\lceil \log_2(L+m) \rceil$ bits in addition to that read-only description.

Current-call arithmetic has separately certified workspace $W_{\mathrm{tmp}}(b)$ and time $T_{\mathrm{eval}}(b)$; at fixed precision its expected total time is at most $T_{\mathrm{eval}}(b)\mathbb{E}\tau$ plus preparation, planning and audit. For the gate example the dependence on p must remain in the potential or exact cycle count. These are constructive upper certificates, not optimal lower bounds for all computational resources.

8. Common-task calibration and causal simulation

A causal experiment morphism consists of a parameter-independent report transformation, strategy lift, internal simulator state, compatible actions, physical clock and task readout. Its deficiency is measured on the complete stopped and scored law, including the observer's fresh coins. If that total variation defect is at most ε and $0 \le \ell \le L_\ell$, then

$$R_{\mathrm{source}}(\rho(B)) \le R_{\mathrm{target}}(B) + L_\ell \varepsilon. \tag{8.1}$$

Indeed run the lifted observer and simulator jointly, compare their bounded common-task expectations, and take an infimum. Two such defects add by contraction and the triangle inequality. If the simulator uses R internal states and the observer uses M labels, their direct product uses at most RM labels. Mean resource transfer requires its own pathwise or potential certificate; it does not follow from total variation. Theorem 7.1 supplies one such certificate with protocol modes that need not stop at each call. A one-way morphism does not imply a converse risk inequality.

If the final prediction differs in root mean square by at most δ_r under the same law, Minkowski's inequality gives

$$\sqrt{\mathbb{E}\|Z - \widehat{r}\|^2} \le \sqrt{\mathbb{E}\|Z - r\|^2} + \delta_r.$$

Root errors add before squaring. An upper tolerance δ_r is not an information-theoretic lower bound. The following attained-error experiment makes the distinction explicit.

Theorem 8.1 (An attained joint resource–risk law)**.** *Use either gate realization with $B_0 = 0$ and fixed calibration/geometric constants. Before the gate protocol, one paid call prepares a fair sign $U \in \{-1, 1\}$ and reports it unless it is erased with probability 2ε, $0 \le \varepsilon \le 1/2$. The core starts independently of U; none of its kernels or audit coordinates reads U. At the final paid audit score the vector $(Z, U, \xi\delta)$, where $\xi \in \{-1, 1\}$ is a fixed unknown calibration alternative with identical pre-audit laws. The sign response is restricted to $[-1, 1]$. Count the early call and final audit, so a total budget N leaves $N - 2$ expected raw calls for the core.*

For $N \ge \kappa(p) + 2$ and $M \ge 13$, put

$$m_M = \left\lfloor \frac{M-1}{3} \right\rfloor - 2 \ge 2, \qquad b = \frac{N-2}{\kappa(p)}.$$

The worst-model autonomous risk for this same task satisfies

$$\mathcal{R}(N, M, \delta, \varepsilon) \asymp F_{d,\alpha}(b, m_M) + \delta^2 + \varepsilon. \tag{8.2}$$

The constants are independent of $p, N, M, \delta, \varepsilon$ for bounded δ and fixed core certificates. The early channel's causal deficiency to an exact immediate sign-report interface is exactly ε. It must complete that virtual report before entering the core.

Proof. The core lower holds after every allowed history, including knowledge of an unerased U and all previous coins. Additional independent randomness cannot defeat its actual-occupation bound. Projecting all terminal values of an M-label program to the core gives at most $M - 1$ core values. Consequently its core risk is at least a constant times $F_{d,\alpha}(b, M-1)$. Conditional on erasure, U remains fair and independent of all permitted reports and actions: this follows inductively because the core kernels do not read U. Its unavoidable squared loss is then one, contributing 2ε. The two calibration alternatives have identical pre-audit laws; averaging their risks for any response r gives $\mathbb{E}r^2 + \delta^2$. Taking the worse alternative yields at least δ^2. These lower terms can be combined on the same task and model; alternatively their maximum bounds a third of their sum.

For the upper use one initial label, three core lanes for $U = +1, -1$, or erased, and in each lane two live protocol labels and m_M terminal labels. The total is $1 + 6 + 3m_M \le M$. Run the common core program in each lane, predict the reported sign or zero on erasure, and predict zero for the calibration coordinate. Its risk is at most a constant times $F_{d,\alpha}(b, m_M) + 2\varepsilon + \delta^2$, and its expected total calls are at most N. Since $m_M \ge 2$ and $M - 1 < 3m_M + 9 \le 8m_M$, the two core widths are comparable uniformly, proving (8.2).

On erasure any simulator completing the sign report immediately has no information about U and errs with probability at least $1/2$. Completion by an independent fair sign attains that error. The complete joint law of true and virtual signs is therefore at total variation distance exactly $(2\varepsilon)/2 = \varepsilon$ from the exact-sign law. Later physical audits cannot be used to complete an earlier virtual report. This proves attained deficiency, not a lower bound inferred from an arbitrary allowed tolerance. □

The total budget in (8.2) concerns a stopped protocol, not a deterministic horizon. Calibration and numerical errors are distinguished: δ in this theorem is an actual hidden alternative, while numerical routing errors require the separate feasible-cost certificates of Section 7. Deterministic sample caps, unbounded losses, large-deviation equivalence and infinite-horizon limit interchange are not consequences of the displayed total variation inequality.

Appendix A. Executable tests and predictive realization

The protocol projection is not a replacement for the predictive quotient. Starting from the prepared kernel, let $\mathcal{T}_i$ be a family of bounded executable future tests closed under legal one-call continuation, finite randomization and the bounded monotone limits being used. Include report events, audit coordinates and the legal action signature. For a visible history h define its evaluation vector $q(h) = (\mathbb{E}[T \mid h])_{T \in \mathcal{T}_i}$. Two histories are equivalent only if their signatures and all these predictions agree. Unknown-model and fixed-prior interpretations are distinct: uniform predictions retain the family indexed by θ, while a Bayesian quotient requires a stated prior on the joint model and hidden state.

Proposition A.1 (Realized predictive quotient)**.** *Assume a countable determining family of tests, Borel evaluation, a Borel attained image Q, and a Borel section of the evaluation map on that image. Suppose one-step report kernels and next evaluations depend only on the current evaluation and compatible action, as verified by the unnormalized test closure. Then Q carries a Borel predictive update and is minimal among Borel statistics that determine the same executable tests and admit the corresponding measurable factorization. This statement does not assert that every measurable quotient is standard Borel.*

Proof. Countably many Borel evaluations form a Borel map into a countable product of bounded intervals. Test closure implies that equivalent histories have the same next-report probabilities and the same normalized next-test values at positive evidence. Select a representative with the assumed Borel section and apply the raw update there. The result is independent of the representative, and the section makes it Borel jointly with the permitted action. Zero-evidence reports do not require an arbitrarily chosen posterior. Induction establishes equality of every finite future test law; the declared monotone closure extends this to the stated test class. Conversely different evaluations are separated by a determining test. A statistic sufficient for that class has fibers contained in evaluation fibers, so the quotient factors through it; measurability uses the factorization condition in the statement. □

In the finite hidden-state and finite Hermitian realizations, unnormalized positive coordinates and explicit finite audit effects give a concrete compact carrier. Quotient directions that no admitted future test separates are removed; visible mode and active audit registers are retained. Constant density or finite-dimensionality alone is not invoked as a measurable-image theorem. If a global

standard Borel quotient has not been established, the full posterior on the stated standard Borel hidden space remains a valid implementation for defining conditional tests, but is not handed to a finite observer for free. Generalization from finite matrices to unbounded operator closures requires additional mathematics.

Appendix B. Dependencies and extent of the theorem

The main proof uses conditional projection, positive finite-flow realization, weak-star compactness on a bounded count slice, and a geometric packing argument. Neither physical realization is used to prove the central theorem. The finite protocol projection is exact for the primary report, not a consequence of mere positivity or a claim that the additional report is neutral. Uniform conditional calibration is verified on raw invariant families; it can fail when accumulated history makes the residual uncertainty smaller than every current-report mark. The live-state lower uses legal action signatures rather than an uncharged phase signal.

The full stopped-experiment companion [4] is retained as a separate mathematical article. It contains the one-context density-envelope theorem, exact mass–cost dual, Gaussian whole-history lower and three-state construction, revealing-mass phase transition, and complete two-state ROC including atoms. Those statements keep their original hypotheses and are not reclassified as newly proved here. Its own complete companion chain preserves earlier predictive, posterior-pooling, continuation, finite-horizon, stationary and stopping results. The present central theorem is self-contained and does not require the reader to accept that entire archive as hidden premises.

The finite protocol class supplies a necessary and sufficient flow condition for baseline risk and matching full profiles when mandatory incompatible modes force the live-state charge. It does not supply a near-necessary classification for all raw causal kernels. General unknown-kernel learning, vanishing hidden-state positivity floors, nonclosed infinite observable drivers, efficient global synthesis and joint optimal lower bounds for program description, workspace, precision and physical time remain separate proof obligations. In particular, allowing $p \downarrow 0$ in a paid finite driver is not a proof of uniformity as the hidden filtering parameter q vanishes. These distinctions concern the proven statements, not a change of the foundational mother problem.

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